Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Fagundes Bros. Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
2510 Turlock RD	Snelling	Merced	95369

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1955

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X200-X047-XXXX

B. OPERATORS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Operator name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Legal owner name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	1,292	138	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	1,292	138	0	0	0	0
Average number	1,292	138	0	0	0	0
Avg live weight (lbs)	1,400	1,500	0	0		

Predominant milk cow breed: Holstein

Average milk production: 61 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 33,812.69 *tons per reporting period*

Total nitrogen from manure: 440,653.39 *lbs per reporting period*

After ammonia losses (30% loss applied): 308,457.37 *lbs per reporting period*

Total phosphorus from manure: 73,400.60 *lbs per reporting period*

Total potassium from manure: 239,790.32 *lbs per reporting period*

Total salt from manure: 640,071.30 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 30,437,240 *gallons*

Total nitrogen generated: 56,059.96 *lbs*

Total phosphorus generated: 10,671.88 *lbs*

Total potassium generated: 78,916.37 *lbs*

Total salt generated: 489,639.00 *lbs*

	30,437,240	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	30,437,240	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - 56-WD-1	Ground water
Domestic Well - 56-WD-2	Ground water
Irrigation Water - Surface Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	35	35	2	both	X042-X200-X047-XXXX
10	30	30	0	none	X042-X200-X047-XXXX
11	20	20	1	both	X042-X200-X047-XXXX
2A	32	32	1	process wastewater	X042-X200-X047-XXXX
2B	27	27	1	process wastewater	X042-X200-X047-XXXX
2C	25	25	1	process wastewater	X042-X200-X047-XXXX
3	80	80	2	both	X042-X200-X047-XXXX
4	50	50	1	both	X042-X170-X010-XXXX
7	80	80	2	both	X042-X200-X047-XXXX
Totals for areas that were used for application	349	349	11		
Totals for areas that were not used for application	30	30	0		
Land application area totals	379	379	11		

B. CROPS AND HARVESTS

1

Field name: 1

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	551.00 *ton*	Dry-weight		72.1	16,200.00	3,900.00	24,000.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.74	142.31	34.26	210.83	641.27

1

08/13/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 35 Plant date: 08/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	359.00 *ton*	Dry-weight		84.6	26,700.00	3,300.00	33,700.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.26	84.35	10.43	106.47	347.51

11

Field name: 11

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	521.00 *ton*	Dry-weight		61.8	13,200.00	2,900.00	13,900.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.05	262.71	57.72	276.64	1,373.25

2A

Field name: 2A

2A

12/04/2022: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 32 Plant date: 12/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	97.00 *ton*	Dry-weight		63.2	30,000.00	2,900.00	25,400.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	3.03	66.93	6.47	56.67	252.10

2B

Field name: 2B

12/04/2022: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 27 Plant date: 12/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	83.00 *ton*	Dry-weight		58.9	31,900.00	2,500.00	18,300.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	3.07	80.61	6.32	46.24	235.00

2C

Field name: 2C

2C

11/25/2019: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 25 Plant date: 11/25/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	76.00 *ton*	Dry-weight		58.3	34,400.00	3,700.00	23,800.00		14.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	3.04	87.22	9.38	60.34	370.16

3

Field name: 3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,242.00 *ton*	Dry-weight		69.9	12,100.00	3,200.00	15,200.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.53	113.09	29.91	142.06	719.65

05/15/2023: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 80 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	240.00 *ton*	Dry-weight		66.3	38,500.00	3,600.00	23,600.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	3.00	77.85	7.28	47.72	226.46

4

Field name: 4

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,532.00 *ton*	Dry-weight		61.0	9,900.00	2,800.00	12,900.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.64	236.60	66.92	308.30	1,338.36

7

Field name: 7

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	1,186.00 *ton*	Dry-weight		70.6	16,800.00	2,100.00	17,900.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.83	146.45	18.31	156.04	740.95

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/07/2023	2,365.00 *ton*	Dry-weight		69.4	12,100.00	3,200.00	14,900.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.56	218.92	57.90	269.57	1,212.18

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/19/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.95	9.60	72.11	450.37	954,000.00 *gal*
Irrigation Water - Surface Water	Surface water	2.53	0.00	0.00	25.77	3,726,700.00 *gal*
Application event totals		53.48	9.60	72.11	476.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.77	12.01	90.24	563.67	1,194,000.00 *gal*
Irrigation Water - Surface Water	Surface water	2.65	0.00	0.00	27.01	3,906,300.00 *gal*
Application event totals		66.42	12.01	90.24	590.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	69.91	19.29	42.36	0.00	210.00 *ton*
Application event totals		69.91	19.29	42.36	0.00	

1 - 08/13/2023: Sudangrass, silage

Field name: 1

Crop: Sudangrass, silage Plant date: 08/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 08/13/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.09	0.00	0.00	31.48	4,552,860.00 *gal*
Application event totals		3.09	0.00	0.00	31.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.76	10.50	78.91	492.86	1,044,000.00 *gal*
Irrigation Water - Surface Water	Surface water	3.08	0.00	0.00	31.36	4,534,900.00 *gal*
Application event totals		58.84	10.50	78.91	524.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.88	10.08	59.50	329.17	1,062,000.00 *gal*
Irrigation Water - Surface Water	Surface water	3.09	0.00	0.00	31.48	4,552,860.00 *gal*
Application event totals		48.98	10.08	59.50	360.65	

11 - 06/03/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	192.26	53.04	116.49	0.00	330.00 *ton*
Application event totals		192.26	53.04	116.49	0.00	

11 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	4.09	0.00	0.00	41.62	3,439,340.00 *gal*
Application event totals		4.09	0.00	0.00	41.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.27	11.54	86.71	541.58	655,540.00 *gal*
Irrigation Water - Surface Water	Surface water	4.07	0.00	0.00	41.40	3,421,380.00 *gal*
Application event totals		65.34	11.54	86.71	582.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.96	0.00	0.00	40.31	3,331,580.00 *gal*
Application event totals		3.96	0.00	0.00	40.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.59	11.23	84.33	526.74	637,580.00 *gal*
Irrigation Water - Surface Water	Surface water	3.81	0.00	0.00	38.79	3,205,860.00 *gal*
Application event totals		63.40	11.23	84.33	565.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.81	0.00	0.00	38.79	3,205,860.00 *gal*
Application event totals		3.81	0.00	0.00	38.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.79	0.00	0.00	38.57	3,187,900.00 *gal*
Application event totals		3.79	0.00	0.00	38.57	

2A - 12/04/2022: Clover-grass, hay

Field name: 2A

Crop: Clover-grass, hay

Plant date: 12/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.47	0.00	0.00	25.13	3,322,600.00 *gal*
Application event totals		2.47	0.00	0.00	25.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.38	0.00	0.00	24.24	3,205,860.00 *gal*
Application event totals		2.38	0.00	0.00	24.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.39	13.26	78.32	433.26	1,278,000.00 *gal*
Irrigation Water - Surface Water	Surface water	2.31	0.00	0.00	23.50	3,107,080.00 *gal*
Application event totals		62.70	13.26	78.32	456.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.32	0.00	0.00	23.57	3,116,060.00 *gal*
Application event totals		2.32	0.00	0.00	23.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.28	0.00	0.00	23.16	3,062,180.00 *gal*
Application event totals		2.28	0.00	0.00	23.16	

2A - 12/04/2022: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.37	0.00	0.00	24.11	3,187,900.00 *gal*
Application event totals		2.37	0.00	0.00	24.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.28	0.00	0.00	23.16	3,062,180.00 *gal*
Application event totals		2.28	0.00	0.00	23.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.28	0.00	0.00	23.16	3,062,180.00 *gal*
Application event totals		2.28	0.00	0.00	23.16	

2B - 12/04/2022: Clover-grass, hay

Field name: 2B

Crop: Clover-grass, hay

Plant date: 12/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.51	0.00	0.00	25.52	2,846,660.00 *gal*
Application event totals		2.51	0.00	0.00	25.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.38	0.00	0.00	24.23	2,702,980.00 *gal*
Application event totals		2.38	0.00	0.00	24.23	

2B - 12/04/2022: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.93	14.87	111.69	697.64	1,140,000.00 *gal*
Irrigation Water - Surface Water	Surface water	2.40	0.00	0.00	24.47	2,729,920.00 *gal*
Application event totals		81.33	14.87	111.69	722.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.32	0.00	0.00	23.58	2,631,140.00 *gal*
Application event totals		2.32	0.00	0.00	23.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.31	0.00	0.00	23.50	2,622,160.00 *gal*
Application event totals		2.31	0.00	0.00	23.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.29	0.00	0.00	23.34	2,604,200.00 *gal*
Application event totals		2.29	0.00	0.00	23.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.21	0.00	0.00	22.46	2,505,420.00 *gal*
Application event totals		2.21	0.00	0.00	22.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.21	0.00	0.00	22.54	2,514,400.00 *gal*
Application event totals		2.21	0.00	0.00	22.54	

2C - 11/25/2019: Clover-grass, hay

Field name: 2C

Crop: Clover-grass, hay

Plant date: 11/25/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.49	0.00	0.00	25.30	2,613,180.00 *gal*
Application event totals		2.49	0.00	0.00	25.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.39	0.00	0.00	24.34	2,514,400.00 *gal*
Application event totals		2.39	0.00	0.00	24.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.69	16.14	121.26	757.42	1,146,000.00 *gal*
Irrigation Water - Surface Water	Surface water	2.38	0.00	0.00	24.25	2,505,420.00 *gal*
Application event totals		88.07	16.14	121.26	781.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.28	0.00	0.00	23.21	2,397,660.00 *gal*
Application event totals		2.28	0.00	0.00	23.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.29	0.00	0.00	23.30	2,406,640.00 *gal*
Application event totals		2.29	0.00	0.00	23.30	

2C - 11/25/2019: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.28	0.00	0.00	23.21	2,397,660.00 *gal*
Application event totals		2.28	0.00	0.00	23.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.30	0.00	0.00	23.38	2,415,620.00 *gal*
Application event totals		2.30	0.00	0.00	23.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.28	0.00	0.00	23.21	2,397,660.00 *gal*
Application event totals		2.28	0.00	0.00	23.21	

3 - 11/01/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.74	11.82	88.79	554.56	2,685,020.00 *gal*
Irrigation Water - Surface Water	Surface water	2.61	0.00	0.00	26.57	8,782,440.00 *gal*
Application event totals		65.35	11.82	88.79	581.13	

3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.01	11.31	84.93	530.45	2,568,280.00 *gal*
Irrigation Water - Surface Water	Surface water	2.64	0.00	0.00	26.87	8,881,220.00 *gal*
Application event totals		62.65	11.31	84.93	557.32	

12/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	21.85	6.03	13.24	0.00	150.00 *ton*
Application event totals		21.85	6.03	13.24	0.00	

3 - 05/15/2023: Clover-grass, hay

Field name: 3

Crop: Clover-grass, hay

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.25	0.00	0.00	22.93	7,579,120.00 *gal*
Application event totals		2.25	0.00	0.00	22.93	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.22	0.00	0.00	22.60	7,471,360.00 *gal*
Application event totals		2.22	0.00	0.00	22.60	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	2.22	0.00	0.00	22.63	7,480,340.00 *gal*
Application event totals		2.22	0.00	0.00	22.63	

3 - 05/15/2023: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.91	15.05	113.09	706.36	3,420,000.00 *gal*
Irrigation Water - Surface Water	Surface water	1.97	0.00	0.00	20.05	6,627,240.00 *gal*
Application event totals		81.88	15.05	113.09	726.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	1.96	0.00	0.00	19.99	6,609,280.00 *gal*
Application event totals		1.96	0.00	0.00	19.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	1.97	0.00	0.00	20.07	6,636,220.00 *gal*
Application event totals		1.97	0.00	0.00	20.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	1.97	0.00	0.00	20.07	6,636,220.00 *gal*
Application event totals		1.97	0.00	0.00	20.07	

4 - 06/03/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	174.78	48.22	105.90	0.00	750.00 *ton*
Application event totals		174.78	48.22	105.90	0.00	

4 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	4.04	0.00	0.00	41.12	8,495,080.00 *gal*
Application event totals		4.04	0.00	0.00	41.12	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	4.04	0.00	0.00	41.07	8,486,100.00 *gal*
Application event totals		4.04	0.00	0.00	41.07	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.09	11.32	85.04	531.19	1,607,420.00 *gal*
Irrigation Water - Surface Water	Surface water	3.92	0.00	0.00	39.86	8,234,660.00 *gal*
Application event totals		64.01	11.32	85.04	571.05	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.76	11.26	84.57	528.22	1,598,440.00 *gal*
Irrigation Water - Surface Water	Surface water	3.87	0.00	0.00	39.42	8,144,860.00 *gal*
Application event totals		63.63	11.26	84.57	567.65	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.81	0.00	0.00	38.77	8,010,160.00 *gal*
Application event totals		3.81	0.00	0.00	38.77	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.77	0.00	0.00	38.38	7,929,340.00 *gal*
Application event totals		3.77	0.00	0.00	38.38	

7 - 11/02/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.31	9.29	69.78	435.86	2,110,300.00 *gal*
Irrigation Water - Surface Water	Surface water	2.61	0.00	0.00	26.51	8,764,480.00 *gal*
Application event totals		51.91	9.29	69.78	462.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.68	9.17	68.89	430.29	2,083,360.00 *gal*
Irrigation Water - Surface Water	Surface water	2.65	0.00	0.00	26.92	8,899,180.00 *gal*
Application event totals		51.33	9.17	68.89	457.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	91.76	25.31	55.60	0.00	630.00 *ton*
Application event totals		91.76	25.31	55.60	0.00	

7 - 06/01/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	145.65	40.18	88.25	0.00	1,000.00 *ton*
Application event totals		145.65	40.18	88.25	0.00	

7 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.55	0.00	0.00	36.10	11,934,420.00 *gal*
Application event totals		3.55	0.00	0.00	36.10	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.69	11.62	87.30	545.29	2,640,120.00 *gal*
Irrigation Water - Surface Water	Surface water	3.55	0.00	0.00	36.13	11,943,400.00 *gal*
Application event totals		65.24	11.62	87.30	581.42	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.55	0.00	0.00	36.13	11,943,400.00 *gal*
Application event totals		3.55	0.00	0.00	36.13	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.06	11.50	86.41	539.72	2,613,180.00 *gal*
Irrigation Water - Surface Water	Surface water	3.53	0.00	0.00	35.94	11,880,540.00 *gal*
Application event totals		64.59	11.50	86.41	575.66	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.54	0.00	0.00	36.02	11,907,480.00 *gal*
Application event totals		3.54	0.00	0.00	36.02	

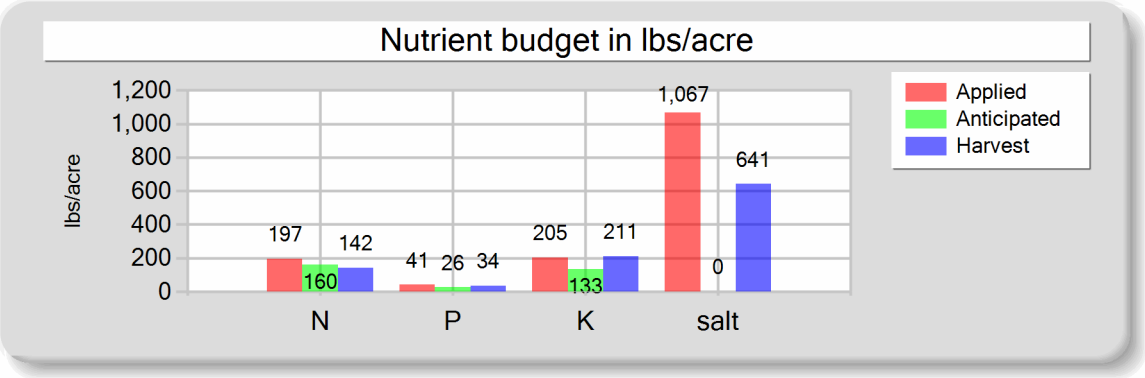
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	3.55	0.00	0.00	36.16	11,952,380.00 *gal*
Application event totals		3.55	0.00	0.00	36.16	

B. NUTRIENT BUDGET

1 - 11/19/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	69.91	19.29	42.36	0.00
Process wastewater	114.72	21.61	162.35	1,014.05
Fresh water	5.19	0.00	0.00	52.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.82	40.90	204.71	1,066.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	142.31	34.26	210.83	641.27
Nutrient balance	54.51	6.64	-6.12	425.55
Applied to removed ratio	1.38	1.19	0.97	1.66

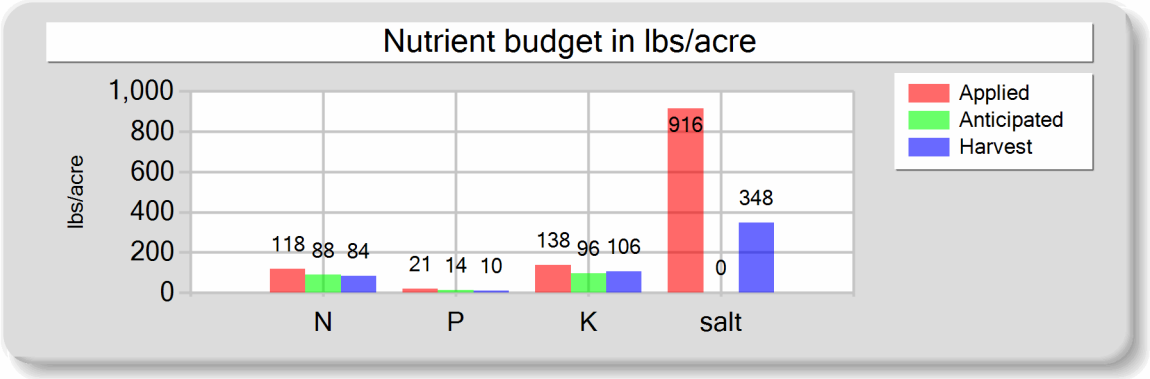
Fresh water applied	
7,633,000.00	*gallons*
281.10	*acre-inches*
8.03	*inches/acre*

Process wastewater applied	
2,148,000.00	*gallons*
79.10	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 08/13/2023: Sudangrass, silage

Field name: 1 Crop: Sudangrass, silage Plant date: 08/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	101.64	20.58	138.41	822.03
Fresh water	9.27	0.00	0.00	94.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	117.91	20.58	138.41	916.35
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	84.35	10.43	106.47	347.51
Nutrient balance	33.56	10.16	31.95	568.84
Applied to removed ratio	1.40	1.97	1.30	2.64

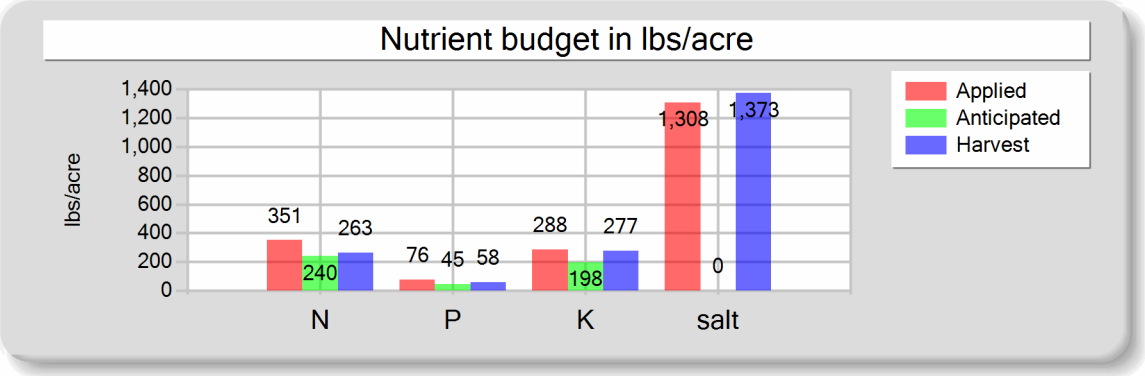
Fresh water applied	
13,640,620.00	*gallons*
502.34	*acre-inches*
14.35	*inches/acre*

Process wastewater applied	
2,106,000.00	*gallons*
77.56	*acre-inches*
2.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 06/03/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	192.26	53.04	116.49	0.00
Process wastewater	120.86	22.77	171.04	1,068.32
Fresh water	23.54	0.00	0.00	239.49
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	350.66	75.81	287.53	1,307.80
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	262.71	57.72	276.64	1,373.25
Nutrient balance	87.95	18.09	10.89	-65.45
Applied to removed ratio	1.33	1.31	1.04	0.95

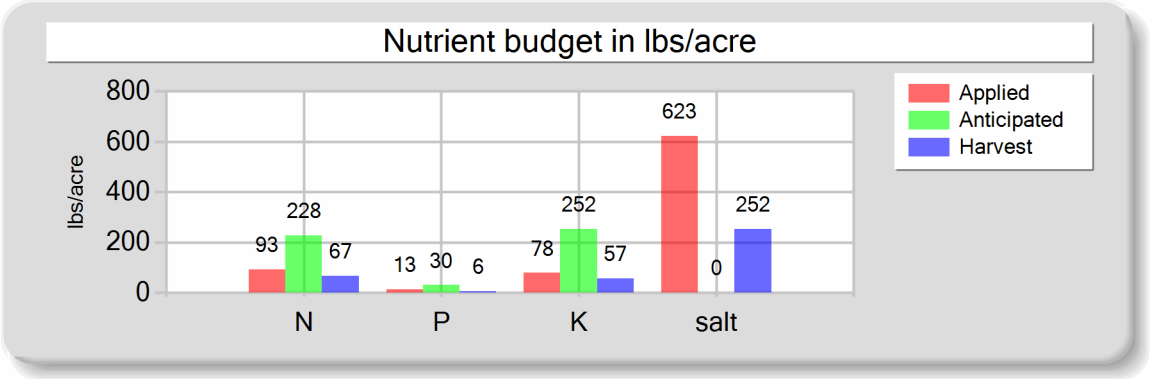
Fresh water applied	
19,791,920.00	*gallons*
728.87	*acre-inches*
36.44	*inches/acre*

Process wastewater applied	
1,293,120.00	*gallons*
47.62	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

2A - 12/04/2022: Clover-grass, hay

Field name: 2A Crop: Clover-grass, hay Plant date: 12/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	60.39	13.26	78.32	433.26
Fresh water	18.67	0.00	0.00	190.02
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	93.06	13.26	78.32	623.28
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	66.93	6.47	56.67	252.10
Nutrient balance	26.13	6.79	21.65	371.18
Applied to removed ratio	1.39	2.05	1.38	2.47

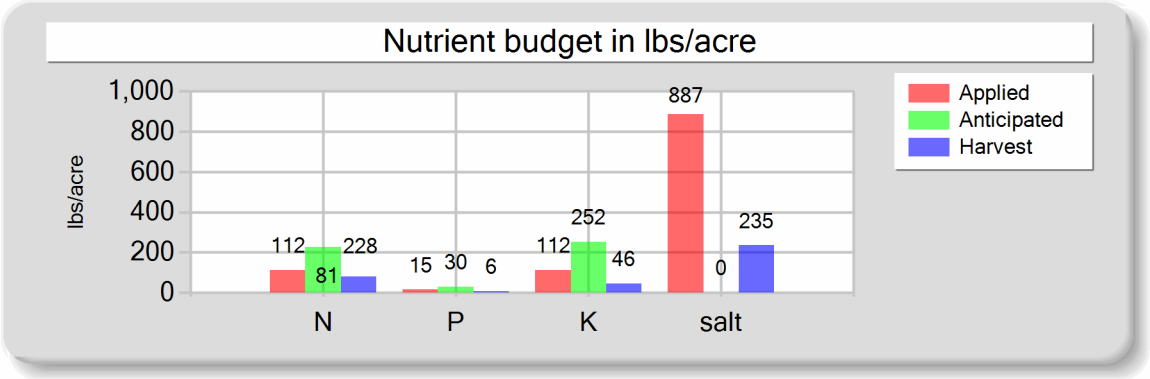
Fresh water applied	
25,126,040.00	*gallons*
925.31	*acre-inches*
28.92	*inches/acre*

Process wastewater applied	
1,278,000.00	*gallons*
47.06	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

2B - 12/04/2022: Clover-grass, hay

Field name: 2B | Crop: Clover-grass, hay | Plant date: 12/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	78.93	14.87	111.69	697.64
Fresh water	18.64	0.00	0.00	189.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	111.56	14.87	111.69	887.27
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	80.61	6.32	46.24	235.00
Nutrient balance	30.95	8.55	65.45	652.27
Applied to removed ratio	1.38	2.35	2.42	3.78

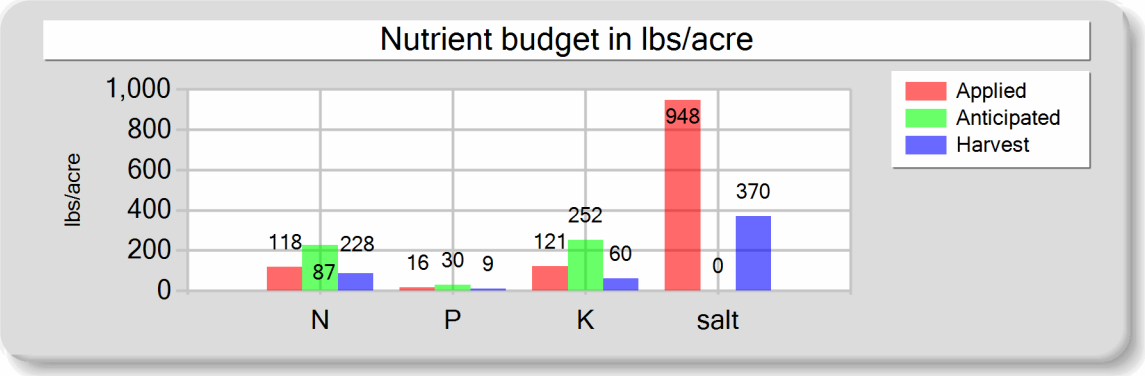
Fresh water applied	
21,156,880.00	*gallons*
779.14	*acre-inches*
28.86	*inches/acre*

Process wastewater applied	
1,140,000.00	*gallons*
41.98	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

2C - 11/25/2019: Clover-grass, hay

Field name: 2C | Crop: Clover-grass, hay | Plant date: 11/25/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	85.69	16.14	121.26	757.42
Fresh water	18.69	0.00	0.00	190.20
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	118.38	16.14	121.26	947.62
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	87.22	9.38	60.34	370.16
Nutrient balance	31.16	6.76	60.92	577.46
Applied to removed ratio	1.36	1.72	2.01	2.56

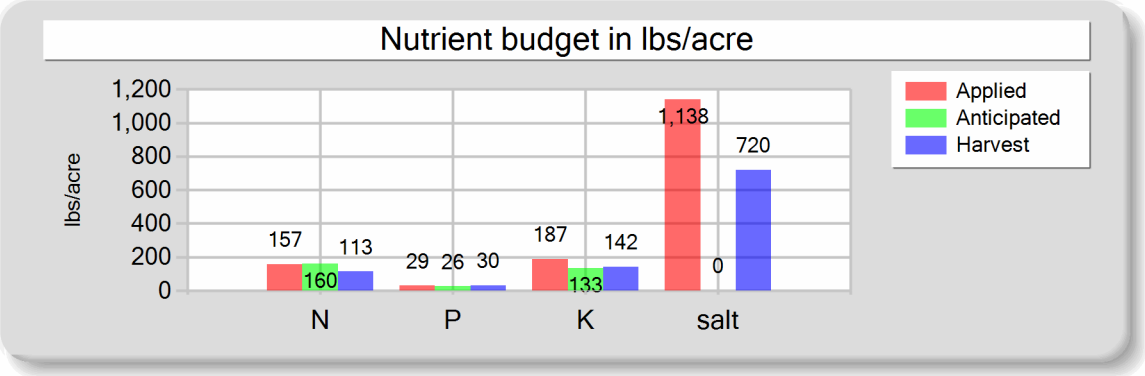
Fresh water applied	
19,648,240.00	*gallons*
723.58	*acre-inches*
28.94	*inches/acre*

Process wastewater applied	
1,146,000.00	*gallons*
42.20	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/01/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	21.85	6.03	13.24	0.00
Process wastewater	122.75	23.12	173.71	1,085.01
Fresh water	5.25	0.00	0.00	53.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.85	29.15	186.95	1,138.44
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	113.09	29.91	142.06	719.65
Nutrient balance	43.76	-0.76	44.89	418.80
Applied to removed ratio	1.39	0.97	1.32	1.58

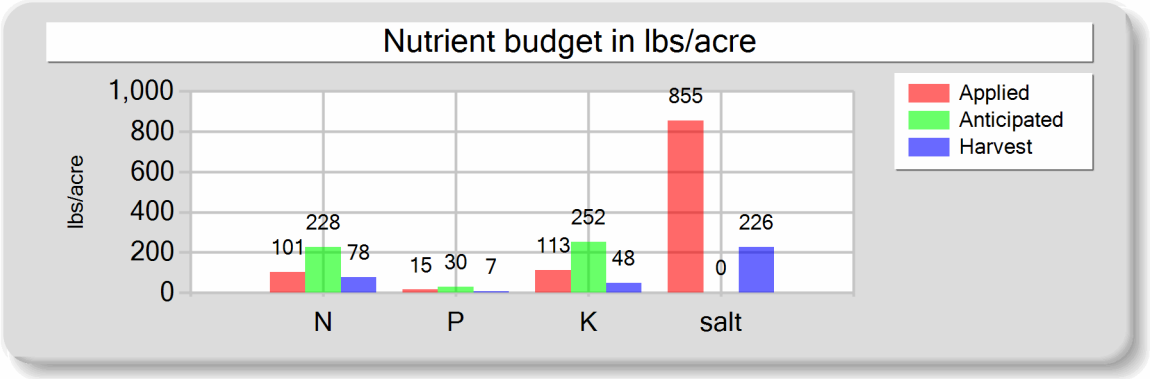
Fresh water applied	
17,663,660.00	*gallons*
650.49	*acre-inches*
8.13	*inches/acre*

Process wastewater applied	
5,253,300.00	*gallons*
193.46	*acre-inches*
2.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/15/2023: Clover-grass, hay

Field name: 3 Crop: Clover-grass, hay Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	79.91	15.05	113.09	706.36
Fresh water	14.58	0.00	0.00	148.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	101.49	15.05	113.09	854.71
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	77.85	7.28	47.72	226.46
Nutrient balance	23.64	7.78	65.37	628.25
Applied to removed ratio	1.30	2.07	2.37	3.77

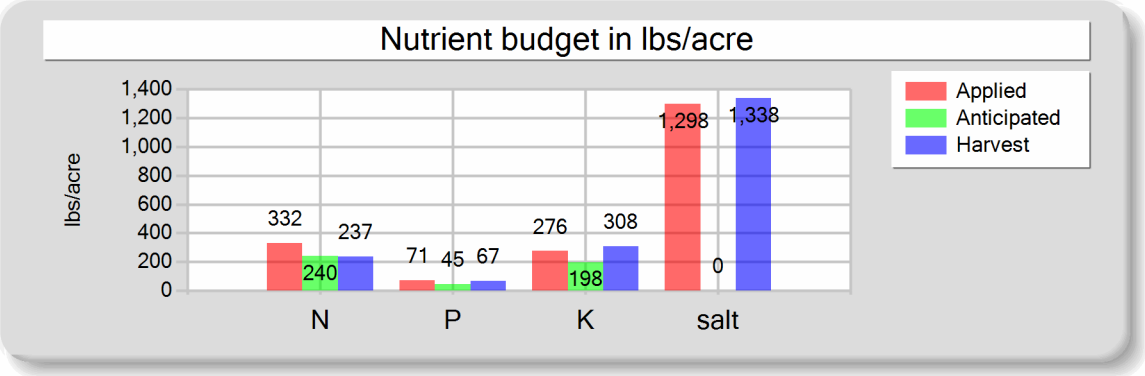
Fresh water applied	
49,039,780.00	*gallons*
1,805.97	*acre-inches*
22.57	*inches/acre*

Process wastewater applied	
3,420,000.00	*gallons*
125.95	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/03/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	174.78	48.22	105.90	0.00
Process wastewater	119.85	22.58	169.61	1,059.41
Fresh water	23.45	0.00	0.00	238.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	332.09	70.80	275.52	1,298.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.60	66.92	308.30	1,338.36
Nutrient balance	95.48	3.88	-32.78	-40.32
Applied to removed ratio	1.40	1.06	0.89	0.97

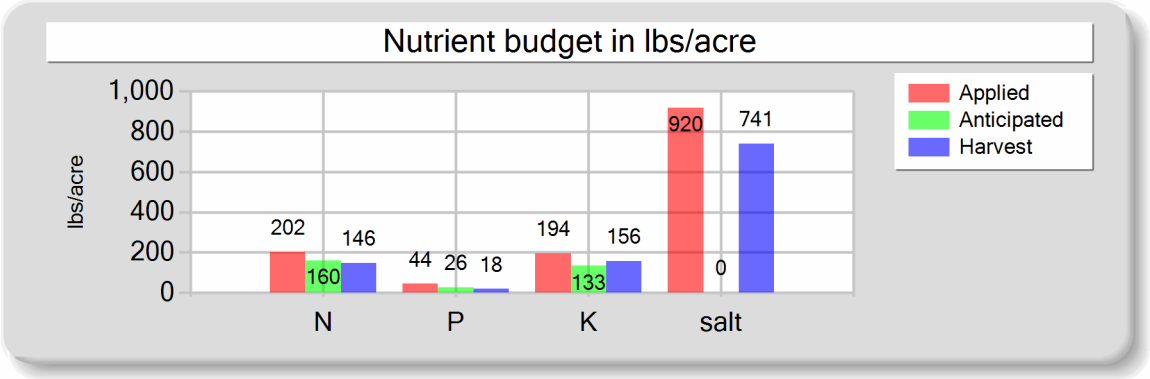
Fresh water applied	
49,300,200.00	*gallons*
1,815.56	*acre-inches*
36.31	*inches/acre*

Process wastewater applied	
3,205,860.00	*gallons*
118.06	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/02/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	91.76	25.31	55.60	0.00
Process wastewater	97.99	18.46	138.67	866.15
Fresh water	5.25	0.00	0.00	53.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.00	43.77	194.27	919.59
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.45	18.31	156.04	740.95
Nutrient balance	55.55	25.47	38.24	178.63
Applied to removed ratio	1.38	2.39	1.25	1.24

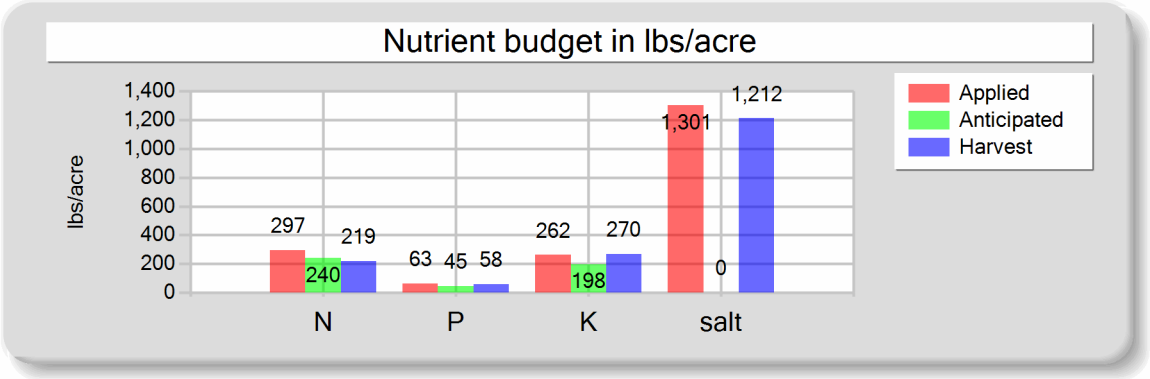
Fresh water applied	
17,663,660.00	*gallons*
650.49	*acre-inches*
8.13	*inches/acre*

Process wastewater applied	
4,193,660.00	*gallons*
154.44	*acre-inches*
1.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/01/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	145.65	40.18	88.25	0.00
Process wastewater	122.75	23.12	173.71	1,085.01
Fresh water	21.27	0.00	0.00	216.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.68	63.30	261.96	1,301.49
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	218.92	57.90	269.57	1,212.18
Nutrient balance	77.76	5.41	-7.61	89.31
Applied to removed ratio	1.36	1.09	0.97	1.07

Fresh water applied	
71,561,620.00	*gallons*
2,635.37	*acre-inches*
32.94	*inches/acre*

Process wastewater applied	
5,253,300.00	*gallons*
193.46	*acre-inches*
2.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid manure

Sample and source description: Solid manure

Sample date: 06/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	5,500.00	7,900.00							0.00
DL	0.01	0.01	0.01							0.01

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	5,600.00	12,300.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 6.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	34.30	8.46	0.00	0.10	4.83	44.90								482.00	320
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	181.00	167.00	0.00	0.20	39.80	235.00								2,980.00	1,300
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	170.00	162.00	0.00	0.30	32.70	229.00								2,590.00	1,340
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	224.00	173.00	0.00	0.00	42.20	317.00								3,340.00	1,980
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - 56-WD-1

Domestic Well - 56-WD-1

56-WD-1

Sample description: 56-WD-1

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00								117.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - 56-WD-2

56-WD-2

Sample description: 56-WD-2

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00								117.00	
DL	0.10	0.10	0.10								10.00	

Irrigation Water - Surface Water

Surface Water

Sample description: Surface Water

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.85	2.82	0.00								27.30	29
DL	1.00	1.00	0.10								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/19/2022: Oats, silage-soft dough

1

Sample and source description: 1

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,900.00	24,000.00		7.30
DL	100.00	100.00	100.00		0.01

1 - 08/13/2023: Sudangrass, silage

1

Sample and source description: 1

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,700.00	3,300.00	33,700.00		11.00
DL	100.00	100.00	100.00		0.01

11 - 06/03/2023: Corn, silage

11 - 06/03/2023: Corn, silage

11

Sample and source description: 11

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,900.00	13,900.00		6.90
DL	100.00	100.00	100.00		0.01

2A - 12/04/2022: Clover-grass, hay

2A

Sample and source description: 2A

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,000.00	2,900.00	25,400.00		11.30
DL	100.00	100.00	100.00		0.01

2B - 12/04/2022: Clover-grass, hay

2B

Sample and source description: 2B

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,900.00	2,500.00	18,300.00		9.30
DL	100.00	100.00	100.00		0.01

2C - 11/25/2019: Clover-grass, hay

2C

Sample and source description: 2C

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,400.00	3,700.00	23,800.00		14.60
DL	100.00	100.00	100.00		0.01

3 - 11/01/2022: Oats, silage-soft dough

3A

Sample and source description: 3A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,200.00	15,200.00		7.70
DL	100.00	100.00	100.00		0.01

3B

Sample and source description: 3B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	3,000.00	14,400.00		6.90
DL	100.00	100.00	100.00		0.01

3 - 05/15/2023: Clover-grass, hay

3 - 05/15/2023: Clover-grass, hay

3A

Sample and source description: 3A

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,500.00	3,600.00	23,600.00		11.20
DL	100.00	100.00	100.00		0.01

3B

Sample and source description: 3B

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,700.00	3,600.00	25,500.00		12.70
DL	100.00	100.00	100.00		0.01

4 - 06/03/2023: Corn, silage

4A

Sample and source description: 4A

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	3,000.00	14,900.00		5.60
DL	100.00	100.00	100.00		0.01

4 - 06/03/2023: Corn, silage

4B

Sample and source description: 4B

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	2,800.00	12,900.00		5.60
DL	100.00	100.00	100.00		0.01

7 - 11/02/2022: Oats, silage-soft dough

7 - A

Sample and source description: 7 - A

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	1,700.00	12,400.00		6.90
DL	100.00	100.00	100.00		0.01

7 - B

Sample and source description: 7 - B

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	2,100.00	17,900.00		8.50
DL	100.00	100.00	100.00		0.01

7 - 06/01/2023: Corn, silage

7 - 06/01/2023: Corn, silage

7-A

Sample and source description: 7-A

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,200.00	14,900.00		6.70
DL	100.00	100.00	100.00		0.01

7-B

Sample and source description: 7-B

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	2,700.00	12,800.00		6.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

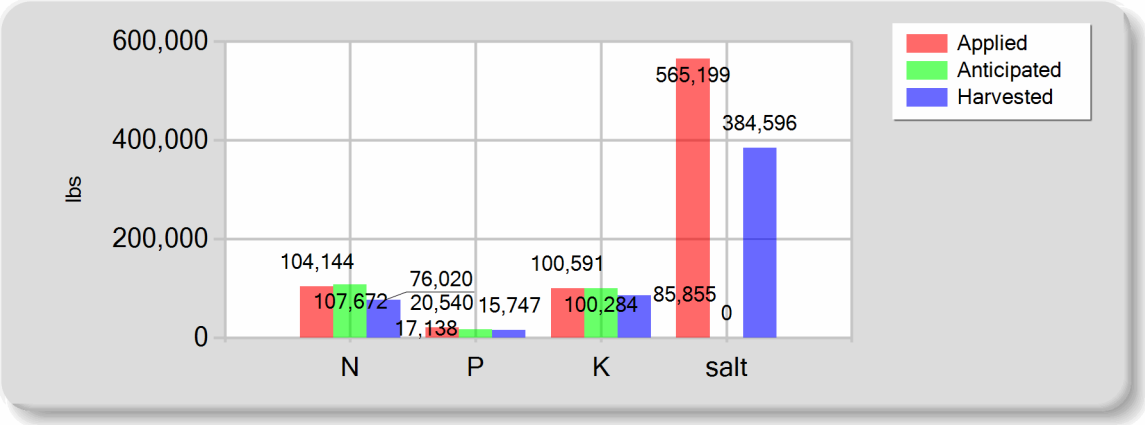
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

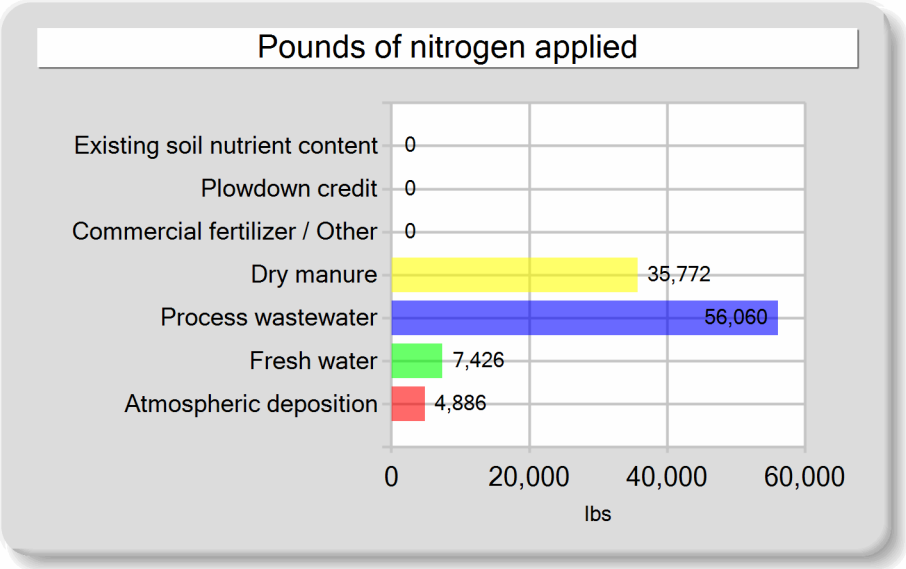
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

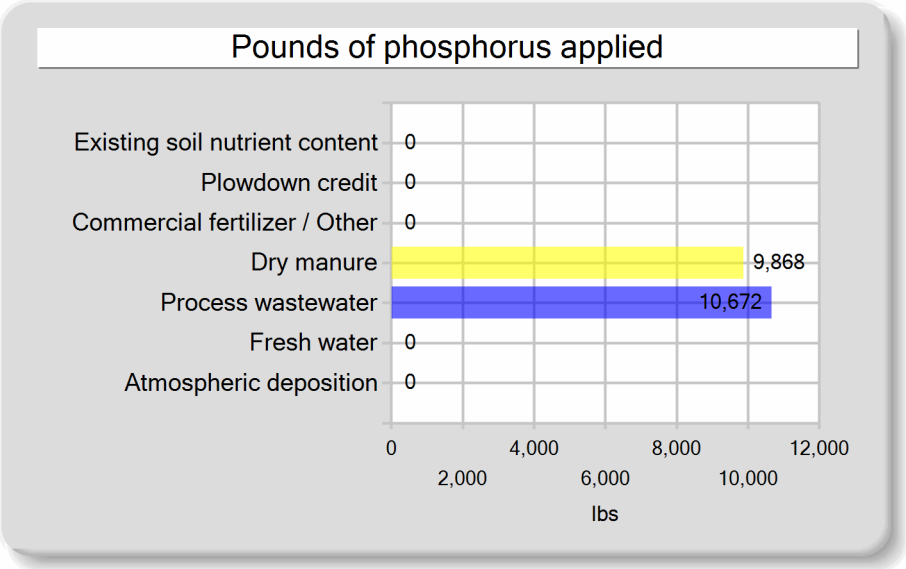
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	35,772.25	9,868.21	21,674.81	0.00
Process wastewater	56,059.96	10,671.88	78,916.37	489,639.00
Fresh water	7,425.74	0.00	0.00	75,560.16
Atmospheric deposition	4,886.00	0.00	0.00	0.00
Total nutrients applied	104,143.95	20,540.09	100,591.18	565,199.16
Anticipated crop nutrient removal	107,672.00	17,138.00	100,284.00	0.00
Actual crop nutrient removal	76,019.77	15,747.33	85,854.68	384,596.06
Nutrient balance	28,124.18	4,792.76	14,736.51	180,603.10
Applied to removed ratio	1.37	1.30	1.17	1.47

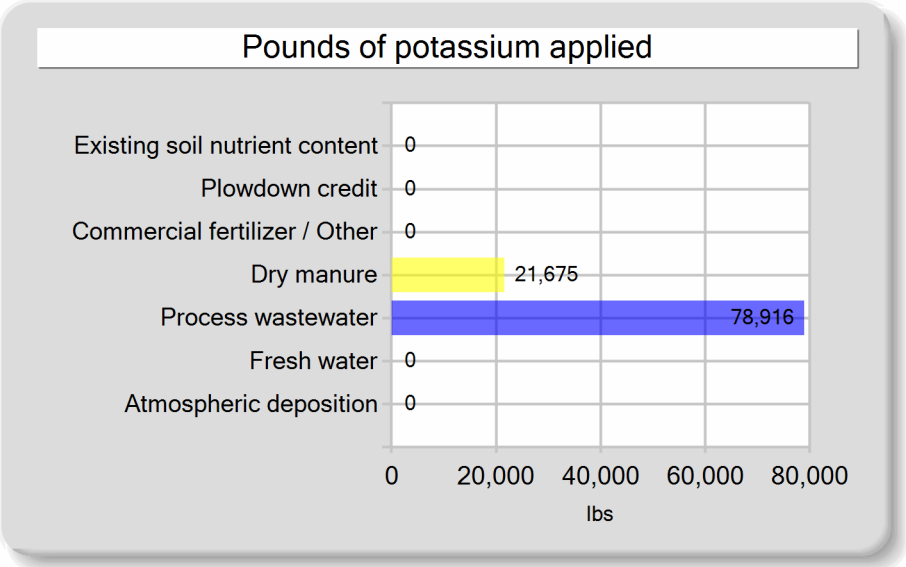
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

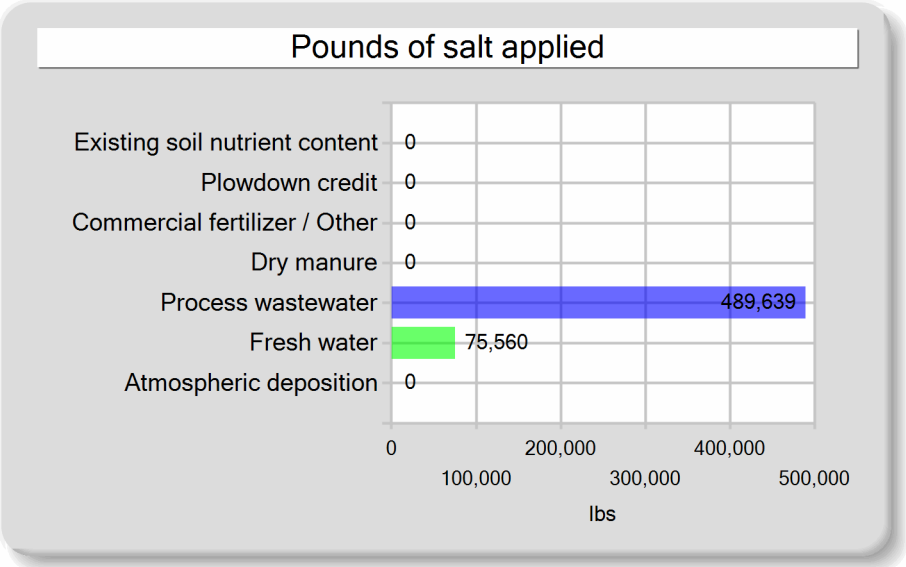


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes
Lloyd Fagundes (Jun 25, 2024 17:05 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Fagundes Bros. Dairy C/O Ralph, Lloyd, &/or Fred Fagundes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/25/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #1

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:55

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0555-01	56-WD-1	Ag Water	T. Madruga		08/02/2023 16:00
23H0555-02	56-WD-2	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: 0.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #1

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:55

Sample Results

Sample: 56-WD-1
23H0555-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.12**	mmhos/cm	0.01	1		08/04/23 14:26	SM 2510 B		BEH0246
Electrical Conductivity umhos	**117**	umhos/cm	10.0	1		08/04/23 14:26	SM 2510 B		BEH0246
Ammonia (as N)	ND	mg/L	0.00	1		08/02/23 16:00	Field		BEH0307
Nitrate Nitrogen as NO3N	**1.0**	mg/L	0.1	1	10	08/03/23 20:40	EPA 300.0		BEH0073
pH	**8.0**	units	1.0	1		08/04/23 14:26	SM 4500-H+	H	BEH0246
Temperature	**25.0**	°C	0.0	1		08/04/23 14:26	SM 2510 B		BEH0246

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LABORATORY INC

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #1

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:55

Sample Results
(Continued)

Sample: 56-WD-2
23H0555-02 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.12**	mmhos/cm	0.01	1		08/04/23 14:27	SM 2510 B		BEH0246
Electrical Conductivity umhos	**116**	umhos/cm	10.0	1		08/04/23 14:27	SM 2510 B		BEH0246
Ammonia (as N)	ND	mg/L	0.00	1		08/02/23 16:00	Field		BEH0307
Nitrate Nitrogen as NO3N	**1.0**	mg/L	0.1	1	10	08/03/23 21:00	EPA 300.0		BEH0073
pH	**8.0**	units	1.0	1		08/04/23 14:27	SM 4500-H+	H	BEH0246
Temperature	**25.0**	°C	0.0	1		08/04/23 14:27	SM 2510 B		BEH0246

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #1

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:55

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0073										
Blank (BEH0073-BLK1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0073-BLK2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0073-BLK3)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0073-BS1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEH0073-BS2)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
Duplicate (BEH0073-DUP1)		**Source: 23H0467-01**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	3.5		0.1	mg/L		3.5			0.482	10
Duplicate (BEH0073-DUP2)		**Source: 23H0555-02**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	1.0		0.1	mg/L		1.0			0.0982	10
Matrix Spike (BEH0073-MS1)		**Source: 23H0467-01**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.5	104	90-110		
Matrix Spike (BEH0073-MS2)		**Source: 23H0555-02**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	6.3		0.1	mg/L	5.000	1.0	105	90-110		
Reference (BEH0073-SRM1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEH0073-SRM2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		
Reference (BEH0073-SRM3)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #1

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:55

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0246										
Blank (BEH0246-BLK1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0246-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0246-BLK3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.7		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEH0246-DUP1)		**Source: 23H0547-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.03		0.01	mmhos/cm		0.03			8.26	10
pH	7.8		1.0	units		7.8			0.383	10
Electrical Conductivity umhos	26.7		10.0	umhos/cm		29.0			8.26	10
Duplicate (BEH0246-DUP2)		**Source: 23H0561-03**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.33		0.01	mmhos/cm		0.32			0.277	10
pH	8.0		1.0	units		8.0			0.251	10
Electrical Conductivity umhos	325		10.0	umhos/cm		324			0.277	10
Reference (BEH0246-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	547			umhos/cm	538.0		102	90-110		
Reference (BEH0246-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.8			units	7.790		101	.7163-101.28		
Reference (BEH0246-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0246-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0246-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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LABORATORY INC

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8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #1

Received: 08/03/2023 15:10
Reported: 08/07/2023 11:55

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0246 (Continued)										
Reference (BEH0246-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0246-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0246-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0246-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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08/03/23 15:10 23H0555

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Fagundes Brothers Dairy #1
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	56-WD-1	8/2/2023	1600	0.5	ND
2	56-WD-2	8/2/2023	1600	-1.8	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JLF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0555

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										